

Weak Convergence: Selected Proofs

MBZUAI: SDS8102 – Advanced Mathematical Statistics

Portmanteau Theorem, Continuous Mapping Theorem,
Uniqueness of Characteristic Functions, and Lévy's Continuity Theorem

Throughout, X_n and X are real-valued random variables with distribution functions F_n and F , respectively.

1. Portmanteau Theorem

Theorem 1.1 (Portmanteau). *The following are equivalent:*

(i) $X_n \xrightarrow{d} X$.

(ii) For every bounded continuous $f : \mathbb{R} \rightarrow \mathbb{R}$,

$$\mathbb{E}[f(X_n)] \rightarrow \mathbb{E}[f(X)].$$

(iii) For every closed $C \subseteq \mathbb{R}$,

$$\limsup_{n \rightarrow \infty} \mathbb{P}(X_n \in C) \leq \mathbb{P}(X \in C).$$

(iv) For every open $G \subseteq \mathbb{R}$,

$$\liminf_{n \rightarrow \infty} \mathbb{P}(X_n \in G) \geq \mathbb{P}(X \in G).$$

Proof. We prove

$$(i) \implies (ii) \implies (iii) \implies (iv) \implies (i).$$

$(i) \implies (ii)$. First suppose that f is continuous with compact support. Choose continuity points $a < b$ of F such that

$$\text{supp}(f) \subset (a, b).$$

For $\varepsilon > 0$, uniform continuity gives a sufficiently fine partition

$$a = x_0 < x_1 < \cdots < x_m = b,$$

whose endpoints are continuity points of F , such that the oscillation of f on each

$$I_j = (x_{j-1}, x_j]$$

is at most ε .

Choose $t_j \in I_j$ and set

$$s(x) = \sum_{j=1}^m f(t_j) \mathbf{1}_{I_j}(x).$$

Then

$$\sup_x |f(x) - s(x)| \leq \varepsilon.$$

Moreover,

$$\mathbb{P}(X_n \in I_j) = F_n(x_j) - F_n(x_{j-1}) \longrightarrow F(x_j) - F(x_{j-1}) = \mathbb{P}(X \in I_j).$$

Hence

$$\mathbb{E}[s(X_n)] \rightarrow \mathbb{E}[s(X)],$$

and therefore

$$\limsup_{n \rightarrow \infty} |\mathbb{E}[f(X_n)] - \mathbb{E}[f(X)]| \leq 2\varepsilon.$$

Letting $\varepsilon \downarrow 0$ proves the claim for compactly supported f .

Now let f be bounded and continuous, with $|f| \leq M$. Choose K such that $\pm K$ are continuity points of F and

$$\mathbb{P}(|X| > K) < \varepsilon.$$

Then

$$\mathbb{P}(|X_n| > K) \rightarrow \mathbb{P}(|X| > K).$$

Let ψ_K be continuous and compactly supported, with

$$0 \leq \psi_K \leq 1, \quad \psi_K = 1 \quad \text{on } [-K, K].$$

Since $f\psi_K$ has compact support,

$$\mathbb{E}[f(X_n)\psi_K(X_n)] \rightarrow \mathbb{E}[f(X)\psi_K(X)].$$

The remaining terms are bounded by

$$M\mathbb{P}(|X_n| > K) \quad \text{and} \quad M\mathbb{P}(|X| > K).$$

Letting $n \rightarrow \infty$ and then $\varepsilon \downarrow 0$ proves

$$\mathbb{E}[f(X_n)] \rightarrow \mathbb{E}[f(X)].$$

(ii) $\Rightarrow$ (iii). Let C be closed and define

$$f_k(x) = \max\{1 - kd(x, C), 0\}, \quad d(x, C) = \inf_{y \in C} |x - y|.$$

Then f_k is bounded and continuous, and

$$\mathbf{1}_C \leq f_k, \quad f_k \downarrow \mathbf{1}_C.$$

Thus

$$\limsup_n \mathbb{P}(X_n \in C) \leq \lim_n \mathbb{E}[f_k(X_n)] = \mathbb{E}[f_k(X)].$$

Letting $k \rightarrow \infty$ and using dominated convergence,

$$\limsup_n \mathbb{P}(X_n \in C) \leq \mathbb{P}(X \in C).$$

(iii) $\Rightarrow$ (iv). If G is open, then G^c is closed. Hence

$$\begin{aligned} \liminf_n \mathbb{P}(X_n \in G) &= 1 - \limsup_n \mathbb{P}(X_n \in G^c) \\ &\geq 1 - \mathbb{P}(X \in G^c) = \mathbb{P}(X \in G). \end{aligned}$$

(iv) $\Rightarrow$ (i). Taking complements first recovers the closed-set inequality in (iii). Let x be a continuity point of F . Since $(-\infty, x]$ is closed,

$$\limsup_n F_n(x) \leq F(x).$$

Since $(-\infty, x)$ is open,

$$\liminf_n F_n(x) \geq \mathbb{P}(X < x) = F(x^-).$$

At a continuity point, $F(x^-) = F(x)$, so

$$F_n(x) \rightarrow F(x).$$

Thus $X_n \xrightarrow{d} X$. ■

2. Continuous Mapping Theorem

Theorem 2.1 (Continuous Mapping Theorem). *Suppose*

$$X_n \xrightarrow{d} X.$$

If $g : \mathbb{R} \rightarrow \mathbb{R}$ is continuous, then

$$g(X_n) \xrightarrow{d} g(X).$$

More generally, the conclusion holds if

$$\mathbb{P}(X \in D_g) = 0,$$

where D_g denotes the set of discontinuity points of g .

Proof. First suppose g is continuous. For every bounded continuous f , the composition $f \circ g$ is bounded and continuous. By Portmanteau,

$$\mathbb{E}[f(g(X_n))] = \mathbb{E}[(f \circ g)(X_n)] \rightarrow \mathbb{E}[(f \circ g)(X)] = \mathbb{E}[f(g(X))].$$

Another application of Portmanteau gives

$$g(X_n) \xrightarrow{d} g(X).$$

For the general statement, let C be closed. Then

$$\overline{g^{-1}(C)} \subseteq g^{-1}(C) \cup D_g.$$

By Portmanteau,

$$\begin{aligned} \limsup_n \mathbb{P}(g(X_n) \in C) &= \limsup_n \mathbb{P}(X_n \in g^{-1}(C)) \\ &\leq \mathbb{P}(X \in \overline{g^{-1}(C)}) \\ &\leq \mathbb{P}(g(X) \in C) + \mathbb{P}(X \in D_g) \\ &= \mathbb{P}(g(X) \in C). \end{aligned}$$

Portmanteau again yields

$$g(X_n) \xrightarrow{d} g(X). \quad \blacksquare$$

3. Moment Generating Functions and Uniqueness

Definition 3.1. The moment generating function (MGF) of a random variable X is

$$M_X(t) = \mathbb{E}[e^{tX}],$$

for those $t \in \mathbb{R}$ for which the expectation is finite.

Unlike characteristic functions, an MGF need not exist away from $t = 0$. However, when it exists on a neighborhood of zero, it uniquely determines the distribution.

Theorem 3.2 (Uniqueness of Moment Generating Functions). *Let X and Y be random variables. Suppose there exists $\delta > 0$ such that*

$$M_X(t) < \infty, \quad M_Y(t) < \infty, \quad |t| < \delta.$$

If

$$M_X(t) = M_Y(t) \quad \text{for every } |t| < \delta,$$

then

$$X \stackrel{d}{=} Y.$$

Proof. Because the MGFs are finite on $(-\delta, \delta)$, they extend to complex arguments

$$z = s + iu, \quad |s| < \delta,$$

through

$$M_X(z) = \mathbb{E}[e^{zX}].$$

Indeed,

$$|e^{zX}| = e^{sX},$$

and the required integrability follows from the existence of the MGF in a neighborhood of zero.

The functions

$$z \mapsto M_X(z), \quad z \mapsto M_Y(z)$$

are analytic on the strip

$$\{z \in \mathbb{C} : |\operatorname{Re} z| < \delta\}.$$

Since they agree on the real interval $(-\delta, \delta)$, the identity theorem for analytic functions implies

$$M_X(z) = M_Y(z)$$

throughout the strip.

In particular, setting $z = iu$ gives

$$\varphi_X(u) = M_X(iu) = M_Y(iu) = \varphi_Y(u) \quad \text{for every } u \in \mathbb{R}.$$

By uniqueness of characteristic functions,

$$X \stackrel{d}{=} Y.$$

■

Important point. It is not enough that all moments exist, or even that

$$\mathbb{E}[X^k] = \mathbb{E}[Y^k] \quad \text{for every } k \geq 0.$$

Different distributions can have the same sequence of moments. Existence of the MGF on an open neighborhood of 0 is a stronger condition that guarantees uniqueness.

MGF versus characteristic function. The two transforms are

$$M_X(t) = \mathbb{E}[e^{tX}], \quad \varphi_X(t) = \mathbb{E}[e^{itX}].$$

The essential distinction is

$$\begin{aligned} M_X(t) &\text{ may fail to exist for } t \neq 0, \\ \varphi_X(t) &\text{ exists for every } t \in \mathbb{R}. \end{aligned}$$

When the MGF exists near zero, it determines the distribution. The characteristic function requires no such integrability assumption and always determines the distribution.

3.1. Do Moments Uniquely Determine a Distribution?

The MGF uniqueness theorem may suggest that knowing all moments

$$\mathbb{E}[X^k], \quad k = 0, 1, 2, \dots,$$

should determine the distribution of X . This is false in general.

A classical counterexample is provided by the lognormal distribution.

Example 3.3 (The lognormal moment problem). *Let*

$$X \sim \text{Lognormal}(0, 1),$$

with density

$$f(x) = \frac{1}{x\sqrt{2\pi}} \exp\left\{-\frac{(\log x)^2}{2}\right\}, \quad x > 0.$$

For $|\varepsilon| \leq 1$, define

$$f_\varepsilon(x) = f(x) [1 + \varepsilon \sin(2\pi \log x)], \quad x > 0.$$

Since

$$1 + \varepsilon \sin(2\pi \log x) \geq 0,$$

these define probability densities. For $\varepsilon \neq 0$, they are different from the original lognormal density.

We claim, however, that all of these distributions have exactly the same moments.

For any integer $k \geq 0$,

$$\int_0^\infty x^k f_\varepsilon(x) dx = \int_0^\infty x^k f(x) dx + \varepsilon \int_0^\infty x^k f(x) \sin(2\pi \log x) dx.$$

It therefore suffices to show that the second term vanishes.

Using the substitution

$$y = \log x,$$

we obtain

$$\int_0^\infty x^k f(x) \sin(2\pi \log x) dx = \frac{1}{\sqrt{2\pi}} \int_{\mathbb{R}} e^{ky - y^2/2} \sin(2\pi y) dy.$$

Completing the square,

$$ky - \frac{y^2}{2} = -\frac{(y-k)^2}{2} + \frac{k^2}{2}.$$

Hence, if $Z \sim N(0, 1)$,

$$\begin{aligned} \int_0^\infty x^k f(x) \sin(2\pi \log x) dx &= e^{k^2/2} \mathbb{E}[\sin(2\pi(Z+k))] \\ &= e^{k^2/2} \mathbb{E}[\sin(2\pi Z)] \\ &= 0, \end{aligned}$$

where we used that k is an integer and symmetry of Z .

Therefore

$$\int_0^\infty x^k f_\varepsilon(x) dx = e^{k^2/2}, \quad k = 0, 1, 2, \dots,$$

independently of ε .

Thus infinitely many distinct distributions have exactly the same sequence of moments.

Why does this not contradict MGF uniqueness? For a lognormal random variable,

$$\mathbb{E}[e^{tX}] = \infty \quad \text{for every } t > 0.$$

Thus its MGF is not finite on any open neighborhood of 0.

The distinction is therefore

$$\text{same moments of every order} \not\Rightarrow \text{same distribution,}$$

whereas

$$\text{same MGF on a neighborhood of 0} \Rightarrow \text{same distribution.}$$

The MGF condition is stronger because, when it exists near zero,

$$M_X(t) = \sum_{k=0}^{\infty} \frac{\mathbb{E}[X^k]}{k!} t^k$$

has positive radius of convergence. In the lognormal example, the moments grow too rapidly:

$$\mathbb{E}[X^k] = e^{k^2/2},$$

and the corresponding formal power series has radius of convergence zero.

4. Characteristic Functions and Uniqueness

Definition 4.1. The characteristic function of a random variable X is

$$\varphi_X(t) = \mathbb{E}[e^{itX}] = \int_{\mathbb{R}} e^{itx} dF_X(x), \quad t \in \mathbb{R}.$$

Since $|e^{itX}| = 1$, the characteristic function always exists. Moreover,

$$\varphi_X(0) = 1, \quad |\varphi_X(t)| \leq 1.$$

If X and Y are independent, then

$$\varphi_{X+Y}(t) = \varphi_X(t)\varphi_Y(t).$$

The fundamental fact is that φ_X determines the entire distribution of X .

Theorem 4.2 (Uniqueness of Characteristic Functions). *If*

$$\varphi_X(t) = \varphi_Y(t) \quad \text{for every } t \in \mathbb{R},$$

then

$$X \stackrel{d}{=} Y.$$

Proof. The idea is to first add a small Gaussian noise. This produces variables with smooth densities, for which ordinary Fourier inversion can be applied.

Let

$$Z \sim N(0, 1)$$

be independent of X and Y . For $\sigma > 0$, define

$$X_\sigma = X + \sigma Z, \quad Y_\sigma = Y + \sigma Z.$$

By independence,

$$\varphi_{X_\sigma}(t) = \varphi_X(t)\varphi_{\sigma Z}(t) = \varphi_X(t)e^{-\sigma^2 t^2/2},$$

and similarly

$$\varphi_{Y_\sigma}(t) = \varphi_Y(t)e^{-\sigma^2 t^2/2}.$$

Since $|\varphi_X(t)| \leq 1$,

$$|\varphi_{X_\sigma}(t)| \leq e^{-\sigma^2 t^2/2},$$

so φ_{X_σ} is integrable. Fourier inversion therefore gives the density

$$p_{X_\sigma}(x) = \frac{1}{2\pi} \int_{\mathbb{R}} e^{-itx} \varphi_X(t) e^{-\sigma^2 t^2/2} dt.$$

Likewise,

$$p_{Y_\sigma}(x) = \frac{1}{2\pi} \int_{\mathbb{R}} e^{-itx} \varphi_Y(t) e^{-\sigma^2 t^2/2} dt.$$

If $\varphi_X = \varphi_Y$, these two expressions are identical. Hence

$$X + \sigma Z \stackrel{d}{=} Y + \sigma Z \quad \text{for every } \sigma > 0.$$

Finally,

$$\sigma Z \xrightarrow{P} 0 \quad \text{as } \sigma \downarrow 0,$$

so

$$X + \sigma Z \xrightarrow{P} X, \quad Y + \sigma Z \xrightarrow{P} Y.$$

Thus

$$X + \sigma Z \xrightarrow{d} X, \quad Y + \sigma Z \xrightarrow{d} Y.$$

Since the distributions on the left are identical for every $\sigma > 0$, their weak limits must also be identical. Therefore

$$X \stackrel{d}{=} Y.$$

■

Interpretation. The Gaussian factor

$$e^{-\sigma^2 t^2/2}$$

makes the characteristic function integrable. This allows Fourier inversion to recover the smoothed distribution. Letting the amount of Gaussian smoothing vanish then recovers the original distribution.

5. Lévy's Continuity Theorem

The uniqueness theorem says that a characteristic function identifies a distribution. Lévy's theorem goes further: convergence of characteristic functions characterizes convergence in distribution.

Theorem 5.1 (Lévy's Continuity Theorem). *Let φ_n be the characteristic function of X_n .*

If

$$X_n \xrightarrow{d} X,$$

then

$$\varphi_n(t) \rightarrow \varphi_X(t) \quad \text{for every } t \in \mathbb{R}.$$

Conversely, suppose

$$\varphi_n(t) \rightarrow \varphi(t) \quad \text{for every } t \in \mathbb{R},$$

and φ is continuous at 0. Then φ is the characteristic function of some random variable X , and

$$X_n \xrightarrow{d} X.$$

Proof. Forward direction. Fix $t \in \mathbb{R}$. The functions

$$x \mapsto \cos(tx), \quad x \mapsto \sin(tx)$$

are bounded and continuous. Hence Portmanteau gives

$$\mathbb{E}[\cos(tX_n)] \rightarrow \mathbb{E}[\cos(tX)]$$

and

$$\mathbb{E}[\sin(tX_n)] \rightarrow \mathbb{E}[\sin(tX)].$$

Therefore

$$\varphi_n(t) = \mathbb{E}[e^{itX_n}] \rightarrow \mathbb{E}[e^{itX}] = \varphi_X(t).$$

Converse direction. The main issue is to show that the distributions of X_n cannot escape to infinity. We first establish tightness.

For $T > 0$, Fubini's theorem gives

$$\begin{aligned} \frac{1}{2T} \int_{-T}^T \varphi_n(t) dt &= \mathbb{E} \left[\frac{1}{2T} \int_{-T}^T e^{itX_n} dt \right] \\ &= \mathbb{E} \left[\frac{\sin(TX_n)}{TX_n} \right], \end{aligned}$$

where the ratio is interpreted as 1 when $X_n = 0$.

Taking real parts,

$$1 - \frac{1}{2T} \int_{-T}^T \operatorname{Re} \varphi_n(t) dt = \mathbb{E} \left[1 - \frac{\sin(TX_n)}{TX_n} \right].$$

There exists a universal constant $c > 0$ such that

$$1 - \frac{\sin y}{y} \geq c \quad \text{whenever } |y| \geq 2.$$

Therefore

$$1 - \frac{\sin(TX_n)}{TX_n} \geq c \mathbf{1}\{|X_n| \geq (2T)^{-1}\},$$

and consequently

$$\mathbb{P}(|X_n| \geq (2T)^{-1}) \leq \frac{1}{c} \left[1 - \frac{1}{2T} \int_{-T}^T \operatorname{Re} \varphi_n(t) dt \right].$$

Now $|\varphi_n(t)| \leq 1$, so dominated convergence gives

$$\frac{1}{2T} \int_{-T}^T \operatorname{Re} \varphi_n(t) dt \longrightarrow \frac{1}{2T} \int_{-T}^T \operatorname{Re} \varphi(t) dt.$$

Also,

$$\varphi(0) = \lim_n \varphi_n(0) = 1.$$

Since φ is continuous at 0,

$$\frac{1}{2T} \int_{-T}^T \operatorname{Re} \varphi(t) dt \longrightarrow 1 \quad \text{as } T \downarrow 0.$$

Hence, for every $\varepsilon > 0$, we can choose T sufficiently small so that

$$\limsup_{n \rightarrow \infty} \mathbb{P}(|X_n| \geq (2T)^{-1}) < \varepsilon.$$

After enlarging T^{-1} if necessary to handle finitely many initial terms, we obtain

$$\sup_n \mathbb{P}(|X_n| > M) < \varepsilon$$

for some $M < \infty$. Thus $\{X_n\}$ is tight.

By tightness, every subsequence $\{X_{n_j}\}$ has a further subsequence $\{X_{n_{j_k}}\}$ such that

$$X_{n_{j_k}} \xrightarrow{d} X$$

for some random variable X .

By the forward direction,

$$\varphi_{n_{j_k}}(t) \rightarrow \varphi_X(t).$$

On the other hand, by assumption,

$$\varphi_{n_{j_k}}(t) \rightarrow \varphi(t).$$

Therefore

$$\varphi_X(t) = \varphi(t) \quad \text{for every } t.$$

Thus φ is itself a characteristic function.

By uniqueness of characteristic functions, every weakly convergent subsequence must have the same limiting distribution. Therefore the whole sequence converges:

$$X_n \xrightarrow{d} X.$$

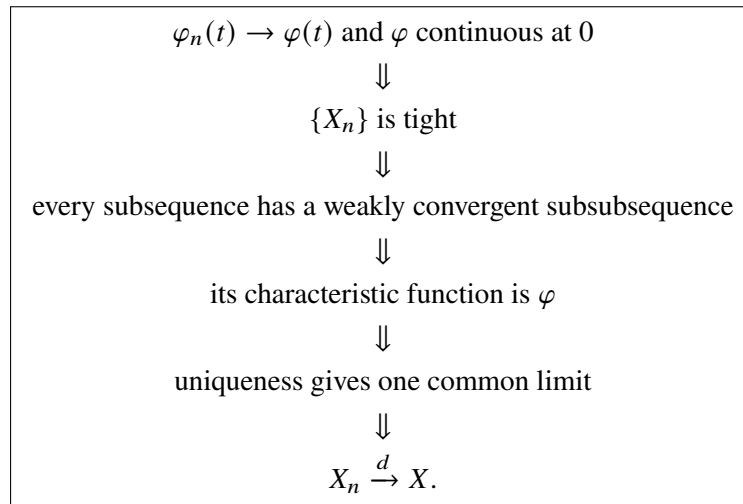
■

Why continuity at zero is needed. Pointwise convergence of characteristic functions alone does not prevent probability mass from escaping to infinity. Continuity of the limit at 0, together with

$$\varphi(0) = 1,$$

is precisely what gives the tightness needed to obtain a limiting probability distribution.

Proof structure. The converse direction can be summarized as



Summary

The main implications established above can be summarized as

$$X_n \xrightarrow{d} X \iff \mathbb{E}[f(X_n)] \rightarrow \mathbb{E}[f(X)] \quad \text{for all bounded continuous } f,$$

and

$$X_n \xrightarrow{d} X \implies g(X_n) \xrightarrow{d} g(X)$$

for continuous g .

Characteristic functions provide an alternative characterization:

$$X_n \xrightarrow{d} X \implies \varphi_{X_n}(t) \rightarrow \varphi_X(t),$$

while Lévy's theorem supplies the converse when the pointwise limit is continuous at the origin.